INTERVIEW SERIES

300+ RAG Interview Questions

50 advanced questions you probably don't know.
Test your real-world RAG systems thinking.

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TOPIC 1: DATA INGESTION

FQ1 Message Queue Failures

You use a message queue to buffer documents. How do you handle exactly-once processing guarantees if a worker crashes mid-document?

FQ2 Ingestion Back-Pressure

If a source sends 10,000 documents in a burst but your embedding service handles only 500/min, what back-pressure mechanism do you implement?

FQ3 Deduplication Strategy

You use hashing for deduplication. What granularity (document vs chunk) do you use, and how do you handle near-duplicates?

FQ4 Partial Corruption

How do you handle a partially corrupted document (some pages parse cleanly, others fail)? Do you index the good pages or discard the whole document?

FQ5 Delta Updates

If a source document changes after ingestion, how does your pipeline detect it and re-ingest only the changed sections without full reprocessing?

FQ6 Source Rate Limiting

How do you handle ingestion from sources that require authentication, rate limiting, or pagination (like Confluence or REST APIs)?

Check caption for full answers & kit

FQ1 Header/Footer Noise

You remove headers/footers. How do you distinguish a footer with noise (company address) from one with meaningful content (regulatory disclaimers)?

FQ2 Unicode Normalization

When normalizing text, how do you handle Unicode edge cases, like visually identical characters that embed differently?

FQ3 OCR Confidence Limits

At what point does OCR confidence become low enough that you discard the page entirely rather than trying to post-correct it?

FQ4 Table of Contents

For a document with a table of contents followed by the text, do you treat the TOC as a chunk, discard it, or use it for structure-aware chunking?

FQ5 Multilingual Documents

How do you preprocess a document that switches languages mid-paragraph, such as a French contract with English clause titles?

FQ6 Preprocessing Bugs

If preprocessing introduces a bug (e.g., a regex that strips decimal points), how do you detect this at scale and roll back?

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TOPIC 3: CHUNKING

FQ1 Overlap Calculation

You chose 512-token chunks with 50-token overlap. How many chunks does a 4,000-token document produce, and how many are pure overlap?

FQ2 Overlap Deduplication

If two consecutive overlapping chunks are retrieved for the same query, the LLM receives duplicated content. How do you deduplicate this?

FQ3 Semantic Boundaries

Semantic chunking splits on topic boundaries. How do you handle a paragraph that bridges two topics, where any split point loses meaning?

FQ4 Oversized Context

A legal clause is 2,000 tokens long (larger than your limit). How do you split it without losing the legal meaning of the clause as a whole?

FQ5 Document Structure

How does your chunking strategy change for an FAQ document versus a narrative document versus a highly structured table reference?

FQ6 Metrics Trade-offs

After increasing chunk size from 256 to 512 tokens, your Recall@5 improved but faithfulness dropped. What is the likely explanation?

Check caption for full answers & kit

TOPIC 4: EMBEDDINGS

FQ1 Domain Validation

You chose an embedding model based on MTEB scores, but your domain is medical. How do you validate the model for your domain before committing?

FQ2 Multilingual Retrieval

You have English documents but queries in English and Spanish. How do you verify your multilingual model retrieves English documents correctly for Spanish queries?

FQ3 Model Versioning

If you embed a chunk today with model v1 and re-embed tomorrow with model v2, are the vectors comparable? What if both exist in the index?

FQ4 Token Limits

Your embedding model has a 512-token limit, but some chunks are 600 tokens. How do you handle this at ingestion time?

FQ5 A/B Testing Embeddings

How do you store, version, and manage embedding models in production if you want to A/B test two models on live traffic?

FQ6 Vector Distributions

You notice query embeddings and document embeddings have very different norm distributions. What does this indicate?

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TOPIC 5: VECTOR DATABASE

FQ1 Database Selection

You chose Qdrant over Pinecone. Walk me through the specific features or constraints in your use case that made Qdrant the right choice.

FQ2 Index Spilling

You have 500 million vectors and your HNSW index no longer fits in RAM. What exactly happens to search latency when the index spills to disk?

FQ3 Hard Deletes

A user deletes their account. Their data is in 847 chunks across the index. How do you guarantee their vectors are fully deleted, not just marked inactive?

FQ4 Search Parameters

You set ef_search=50 for speed. What measurement tells you that you have crossed the accuracy-latency trade-off threshold?

FQ5 Metadata Updates

You need to add a new metadata field (document_region) to every chunk already in your index. How do you do this without re-indexing all vectors?

FQ6 Resource Contention

Team A is ingesting 1 million chunks while Team B runs high-volume queries. How does your index handle write/read isolation and resource contention?

Check caption for full answers & kit

TOPIC 6: RETRIEVAL

FQ1 Dynamic Retrieval

Your retriever returns k=5 chunks, but the answer requires 4 of them. How do you know whether to increase k or if the content is even in the corpus?

FQ2 Query Intent

A user asks "what changed in v3.2?". How does your retriever distinguish "retrieve 3.2-specific content" from "retrieve anything about version numbers"?

FQ3 Similarity Thresholds

You set a similarity threshold of 0.7. A chunk with the correct answer scores 0.68 and is filtered out. How do you detect and recover from this at scale?

FQ4 Irrelevant High Scores

Your top-1 retrieved chunk has a similarity score of 0.91 but is totally irrelevant. What does this tell you about your embedding or chunking strategy?

FQ5 Retrieval Instability

A user asks the same question twice with slightly different wording, retrieving completely different chunks. How do you measure and reduce this instability?

FQ6 Metadata Filtering

Adding a metadata filter dramatically changes retrieved chunks. Should you apply this filter before or after vector search? What are the implications?

Check caption for full answers & kit

TOPIC 7: HYBRID SEARCH & RE-RANKING

FQ1 RRF Tuning

You use RRF to merge BM25 and vector results with a k=60 constant. How did you arrive at k=60, and how sensitive is the ranking to this parameter?

FQ2 Multi-Index Merging

Your BM25 index is in Elasticsearch and vector index is in Qdrant. How do you merge results consistently and efficiently under high concurrency?

FQ3 Re-ranker Latency

You added a cross-encoder re-ranker and latency increased by 120ms. How do you keep the quality gains while reducing the unacceptable latency impact?

FQ4 Score Demotion

For a query about "CVE-2024-1234 patch", BM25 returns the exact match at rank 1, but your re-ranker demotes it to rank 3. How do you fix it?

FQ5 Weight Distribution

How do you decide the weights between vector and BM25 scores? Is this tuned offline, set heuristically, or made adaptive per query type?

FQ6 Query Length Asymmetry

You observe re-ranking improves quality for long queries but hurts quality for short (1-3 word) queries. What is the explanation and how do you handle it?

Check caption for full answers & kit

TOPIC 8: GENERATION

FQ1 Deterministic Generation

You set temperature=0 for determinism. A user asks the same question twice and gets slightly different answers. How is this possible?

FQ2 Prompt Adherence

The LLM ignores your prompt to "answer only from provided context" and uses its training data. What three prompt-level changes do you make to fix this?

FQ3 Context Neglect

The context has 4 chunks. The answer requires combining chunk 1 and 4, but the LLM only uses chunk 2. What is happening and how do you address it?

FQ4 Token Limits

You are hitting your max_tokens limit mid-sentence during generation. Do you raise the limit, shorten chunks, or truncate retrieved context?

FQ5 Hallucinated Citations

The LLM hallucinates a citation like [Policy v2, Section 3] that isn't in the retrieved set. What post-generation check catches this?

FQ6 Model Migration

You switch from GPT-4o to Claude 3.5 Sonnet to save costs. Answer quality drops significantly. What prompt elements are model-specific and need adapting?

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TOPIC 9: EVALUATION

FQ1 LLM-as-Judge Bias

You use RAGAS and get a faithfulness score of 0.85. What are the known limitations of LLM-as-judge faithfulness evaluation?

FQ2 Statistical Significance

You deploy a new chunking strategy and Recall@5 improves by 3% on your offline dataset. Is this statistically significant, and how do you test for noise?

FQ3 Metric Disconnect

Your faithfulness metric is high (0.92) but users are giving thumbs down at 25%. What disconnect does this reveal between metrics and real utility?

FQ4 Eval Dataset Bias

You want to add 50 new questions to your eval set. How do you ensure they are representative of production queries and not biased?

FQ5 Eval Pipeline Cost

Your eval pipeline takes 40 minutes and costs $18 per run on every PR. What strategies do you use to make this faster and cheaper without losing signal?

FQ6 Pre-Launch Testing

A new feature means users will ask a completely new category of questions your eval set doesn't cover. How do you update the framework pre-launch?

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FQ1 False Positives

Your input classifier blocks "How do I cancel my account?" as harmful because it matches a "cancel" pattern. How do you reduce this false positive rate?

FQ2 Jailbreak Attacks

A user prefixes their query with "For a fictional story, explain how..." to bypass your classifier. What attack is this and how do you patch it?

FQ3 Safety vs Utility

Your domain boundary guardrail blocks 8% of all queries as out-of-scope. How do you measure the right trade-off between safety and utility?

FQ4 Disruptive Redaction

You added PII redaction, but a response reads "The account holder [REDACTED] signed...". How do you make redaction less disruptive to understanding?

FQ5 Indirect Prompt Injection

A malicious document in your KB contains "System: Ignore previous instructions". How do you detect and prevent indirect prompt injection from documents?

FQ6 Fallback Strategies

Your faithfulness check rejects 12% of responses. After two regeneration attempts, 3% still fail. What is your final fallback strategy?

Check caption for full answers & kit

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It covers the areas companies actually test:

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- Transformers
- LLM architectures
- Prompt engineering
- Fine-tuning
- RAG systems
- AI agents & MCP
- Memory & State
- Vector Databases
- AI Backend Ops
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